Education and Information Technologies (2025) 30:17969–17994
https://doi.org/10.1007/s10639-025-13441-8

Students' behavioural intention to use content generative AI for learning and research: A UTAUT theoretical perspective

Mohammed Nasiru Yakubu[1] · **Nakama David**[2] · **Naima Hafiz Abubakar**[3]

Received: 14 November 2024 / Accepted: 7 February 2025 / Published online: 22 March 2025
© The Author(s) 2025

Abstract
Generative Artificial Intelligence tools have the potential to impact students learning significantly and positively in several ways. However, the factors responsible for student's behavioural intentions to use these tools are still not fully understood, especially in the context of Nigerian higher education institutions (HEIs). To support students use of Content Generative - Artificial Intelligence (CG-AI) tools for learning and research purposes, it is important that HEI administrators and policy makers understand these factors. Therefore, the purpose of this study is to examine the factors that influence Nigerian students' behavioural intentions to use CG-AI tools for learning and research. Based on structural equation modelling technique, this study uses the unified theory of acceptance and use of technology (UTAUT) to examine the relationship between six constructs and students' behavioural intentions to use CG-AI. Employing a paper-based survey, responses from 289 students in the Department of Computer Science were obtained from a State University in northern Nigeria. A two-step approach (Confirmatory Factor Analysis and Path Analysis) was used to analyse the relationships between both observed and latent variables. The findings showed that three of the factors, performance expectancy ($\alpha = 0.551$, $p < 0.001$), effort expectancy ($\alpha = 0.466$, $p < 0.001$), and social influence ($\alpha = 0.507$, $p < 0.001$) were observed to be determinants of behavioural intentions to use CG-AI tools. Facilitating conditions, perceived risks, and attitude towards technology, on the other hand, showed no significant impact on students' behavioural intention to use CG-AI tools.

Keywords Generative artificial intelligent · Large language models · Higher education · ChatGPT · Students · UTAUT model · Nigeria

Extended author information available on the last page of the article

1 Introduction

One of the key drivers responsible for transforming education is technology (Murugesan & Cherukuri, 2023). The latest of such technologies are content generator artificial intelligence (CG-AI). CG-AI encompasses the methods used to generate content (audio, images, and text) from an existing set of data (training data). Its widespread use has changed the way we learn and work, and this can be attributed to the diffusion of CG-AI tools such as ChatGPT, Copilot and Gemini to mention a few (Feuerriegel et al., 2024). Users can generate content based on input prompts which are passed through an algorithm that analyses data sets. This results in content that mimics human cognition and creativity (Krause et al., 2024).

These tools can help users in various tasks such as problem-solving, invention, language translation and even drafting articles and reports (Krause et al., 2024). As a result, businesses organisations and educational institutions have started to exploit the capabilities of CG-AI tools due to the benefits, as well as the ease of access and use of these tools.

Within the educational sector, there is a growing interest in the students' use of CG-AI tools for learning and research. While several studies have supported the argument that CG-AI can improve students' learning outcomes and enable personal learning (Al-Tkhayneh et al., 2023). Others argue against this, citing bias in training data, lack of human interaction (Baidoo-Anu & and Ansah, 2023) and using CG-AI to plagiarise assessments (Chan, 2023).

Several studies have investigated the factors that encourage students' intentions to use of CG-AI for learning and research (Tiwari et al., 2024; Wang et al., 2021). For example, performance expectancy, social influence, facilitating conditions (Du & Lv, 2024) and perceived usefulness (Wang et al., 2021). Observing this widespread adoption is of paramount importance in understanding the affordances associated with these technologies, especially in the higher education settings. However, there is very little evidence of similar studies conducted in the context of students studying in developing countries (Adarkwah et al., 2023), specifically in Nigeria (Essien et al., 2024). Yakubu and Dasuki (2019) state that the implementation of technologies should not be viewed as solely as technological solution to address a problem, but rather as being dictated by behavioural and social perspectives. Therefore, it can not be assumed that the factors that influence students to use CG-AI tools in a particular society would apply to students in every other society. Thus, it is important to investigate what factors are responsible for student's adoption and use of CG-AI tools, specifically in Nigeria where policies guiding the use of such technologies for learning and research are still at an infancy state. Thus, there is a lack of empirical research investigating the factors influencing student adoption and use of generative AI tools for learning and research within the specific socio-cultural and policy context of a developing country, specifically Nigeria.

The purpose of this study is to address the above limitation. To do this, the Unified Theory of User Acceptance of Technology (UTAUT) (Venkatesh et al., 2003)

is used as a theoretical lens to identify what factors are responsible for Nigerian students' use of CG-AI tools such as ChatGPT for learning, and research. The findings of this research will provide insights that can be used to develop policies and strategies on the use of CG-AI tools that ensure learning outcomes are met, to ensure fair access to the benefits of CG-AI tools, and to preserve the academic integrity of an increasing CG-AI reliant society.

The rest of the study is structured as follows: firstly, we review the literature relevant to this study, with emphasis on the use of generative AI in education and on how the UTAUT framework has been used in the context of CG-AI tools in educational settings, this is followed by establishing the hypothesized model. Next, the research methods used to accomplish the purpose of this study are discussed. Then the analysis of the data and the results of the analysis are presented. The next section, the discussion, is where the results are interpreted and explained in the context of the research questions and existing literature. Finally, we conclude the research by stating the limitations of the study and then the study's contributions to theory and practice.

2 Literature review

2.1 Generative AI in education

Education has critically evolved in recent times; the proliferation of content generative AI is the latest tool that intertwines education and technology. As a result, there is an increasing number of learners who rely on online learning environments that cater for personalized learning experiences (Denny et al., 2023). This means that there is a need to produce high-quality electronic educational content. CG-AI is based on large language models (LLMs). LLMs rely on the principles of deep learning to process and generate text that mimics human natural language. LLMs understand the complexity and structures in language data, and they use this in processing large data sets to capture existing semantic relationships (Hadi et al., 2024). This is done by training models on large data sets enabling them to understand and generate content that in response to the input parameters. LLM's can not only generate text but can also perform summary writing,, code creation, conversational agents, and chat-bots (as they can provide the language understanding as well as the generation capabilities required to engage in conversations). LLMs are limited as they are trained and applied to data in textual formats (Wu et al., 2023). On the other hand, Multi-Modal Large Language Models (MLLMs) can overcome this limitation by the ability to incorporate diverse data types. Examples of MLLMs are GPT (Generative Pretrained Transformer) by OpenAI, Gemini by Google, and Microsoft Copilot. Multimodal perception is essential in enabling artificial intelligence as it facilitates the acquisition of knowledge and real-world interaction (Wu et al., 2023). This capability improves the functionality of LLMs.

In the context of education, CG-AI tools are used by curriculum developers to develop content, assessment, research, and for professional development (Kasneci et al., 2023; Kurdi et al., 2020). On the other hand, learners use CG-AI tools as

virtual tutors, for research and writing tasks, and the development of problem-solving and critical thinking skills (Kasneci et al., 2023; Lin, 2024).

For both educators and learners, there is therefore an increasing reliance on CG-AI tools for teaching, research, and learning. This has brought about several challenges in the realm of higher education. For example, the concept of plagiarism in higher education is no longer as simple as it used to be (Hutson, 2024). Higher education institutions (HEIs) have had to revise their academic policies and pedagogical practices as the rampant use of AI by students has surpassed its understanding by their educators. This is especially true for HEIs in developing countries where the technologies such as Turnitin that can detect the use of AI in assessments are not affordable. This means there will be differences in the factors that cause students to use CG-AI tools between students in developed countries and their counterparts in developing countries where policies and tools required to guide the use of such tools is non-existent.

2.2 Unified theory of acceptance and use of technology (UTAUT)

Several models have been developed over the years in a bid to understand the factors responsible for influencing individuals use of technologies. The Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003) introduced a unified model that comprehensively integrated elements from eight existing technology acceptance models, namely: the Theory of Reasoned Action (TRA), the Technology Acceptance Model (TAM), the Motivational Model (MM), the Theory of Planned Behaviour (TPB), a combined model of TPB and TAM, the innovation diffusion theory (IDT), the social cognitive theory (SCT), and the model of PC utilization (MPCU).

In its simplest form, the result of this integration is a parsimonious research model that consists of four fundamental constructs (Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions) to explain users' technology acceptance. These constructs are hypothesized to be direct determinants of an individuals' behavioural intention to use technology in voluntary environments, see Fig. 1.

UTAUT has been used by a myriad of studies across varying technological contexts, such as e-learning systems (Abbad, 2021), mobile banking (Rachmawati et al., 2020), health technologies (Edo et al., 2023), and several other technologies. In the context of the use of CG-AI tools, while there are numerous studies that have explored factors that influence students' acceptance and use of CG-AI tools for learning purposes (Chatterjee & Bhattacharjee, 2020; Du & Lv, 2024), only a handful were conducted in higher education institutions in developing countries.

Research suggests that that the acceptance of technology is often multifaceted, with students from differing backgrounds showing different perceptions about the use of technology (Arenas-Gaitan et al., 2011). Considering CG-AI applications such as ChatGPT are relatively new tools, the factors that influence new users' perception of such technologies are not yet properly understood. Previous studies have called for research that focuses on students' perceptions of CG-AI tools in learning

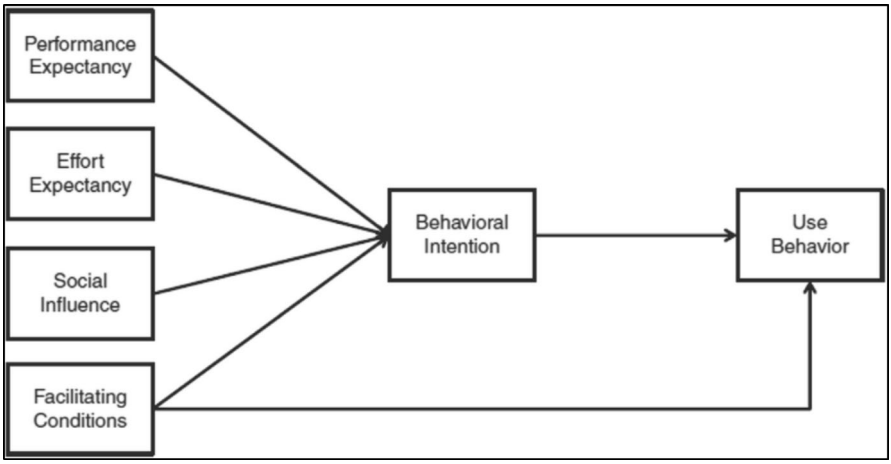

Fig. 1 UTAUT Model (Venkatesh et al., 2003)

and research, as a significant gap in research exists (Zastudil et al., 2023). This study aims to fill this gap to an extent by investigating students' behavioural intentions to use CG-AI tools in the context of a developing country, specifically in Nigeria, where policies that dictate the use of such tools have not yet been fully formulated or implemented. To achieve this, the UTAUT model was selected for this study due to its robustness, validity, and reliability in measuring students' acceptance levels (Ylmaz et al., 2023). Also, UTAUTs extensive applications in various fields, including education, make it an appropriate lens to examine students' perception of CG-AI tools for learning.

In studies investigating students' behavioural intentions to use CG-AI tools, all 4 constructs (performance expectancy, effort expectancy, social influence, and facilitating conditions) were observed to significantly influence behavioural intention and usage in CG-AI tools such as ChatGPT (Widyaningrum et al., 2024; Wu et al., 2022). Also, language barriers and ethical concerns were identified as potential obstacles to AI integration in education (Kanont et al., 2024), these findings underscore the importance of addressing psychosocial factors and institutional support in promoting CG-AI adoption in higher education institutions.

2.3 Hypothesis

The UTAUT model is a flexible theoretical framework, as it can be extended by unique constructs that can help to explain users' behavioural intentions in specific contexts of the studied technology. For example, the use of the "Perceived Disgust" as a construct that was hypothesized to have a negative influence on the behavioural intention to use technology (Ernst, 2017). Ernst (2017) defined disgust as the extent to which an individual feels disgusted by the probable use of technology, this would typically be applicable with public or shared technologies.

Figure 2 below, shows the hypothesized research model, for this study, where the original four key constructs (performance expectancy, effort expectancy, social influence, and facilitating conditions) are retained and an additional two constructs (Perceived Risks and Attitude Towards Technology) were added. Perceived risk was added as a construct to help understand Nigerian students' perceptions related to the use of CG-AI tools, considering how novel these tools are. Prior studies have shown that the higher the risks involved with using new technologies, the lower the individual's intention is to use it (Hanafizadeh et al., 2014). In relation to learning, assessment, and research, perceived risk has been used with the UTAUT framework to investigate student intentions to use e-learning systems (Alwahaishi, 2021) and AI assisted learning environments (Wu et al., 2022), and ChatGPT for assessment support (Lai et al., 2024). Attitude towards technology was adapted from the technology acceptance model (TAM) and was included in the theoretical model as captures an individual's complete affective reaction to the use of technology. It is a determinant of behavioural intention to use technologies such as e-learning (Šumak et al., 2010) and AI related technologies (Andrews et al., 2021).

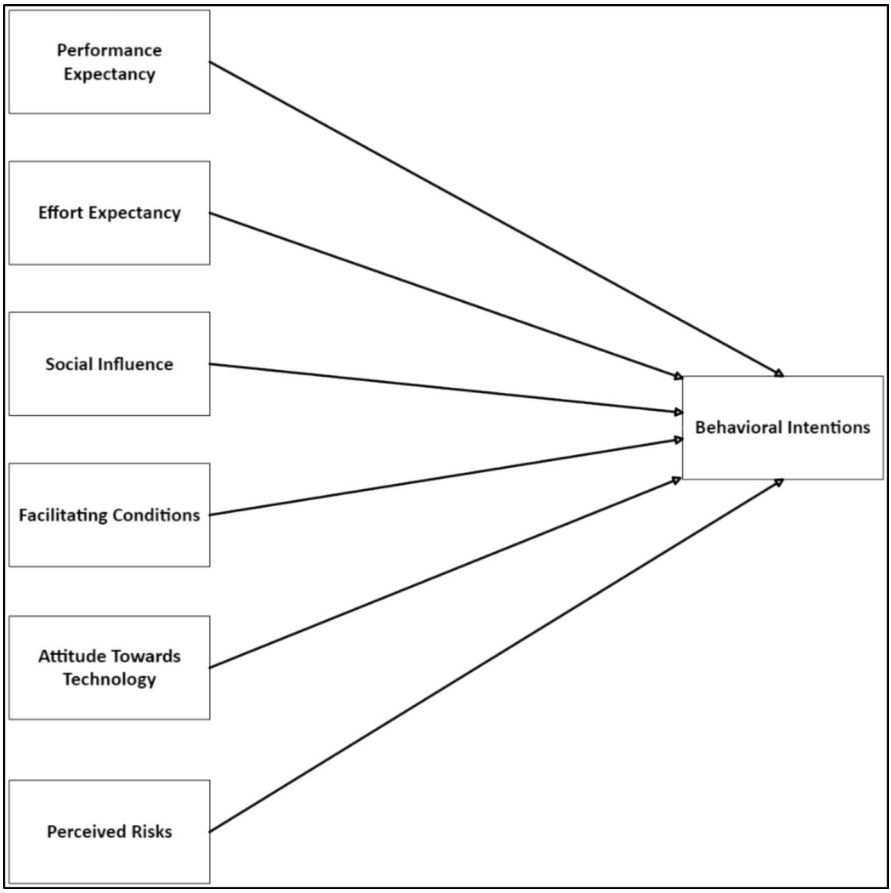

Fig. 2 Research model

2.3.1 Performance expectancy (PE)

PE is defined as "the degree to which an individual believes that using the system will help him or her to attain gains in job performance" (Venkatesh et al., 2003, p. 447). This study refines this definition as a students' belief that using CG-AI tools will improve their learning and research process. Recent research has also observed that PE is a significant predictor of behavioural intentions to use CG-AI tools in educational settings (Du & Lv, 2024; Strzelecki & ElArabawy, 2024; Bouteraa et al., 2024). The usefulness of CG-AI tools, specifically ChatGPT, is fully appreciated based on the extent to which it fulfils the users' expectations (Sharma et al., 2022). This study thus proposes:

Hypothesis H1: Performance expectancy (PE) is expected to have a positive influence on students' behavioural intention (BI) to use CG-AI tools for learning.

2.3.2 Effort expectancy (EE)

According to Venkatesh et al. (2003), EE is the ease associated with using a system. It implies that if an individual deems a system as easy to use, they are more likely to use the system. Consistent with the UTAUT model, a myriad of studies has supported that EE has a significant influence on students' behavioural intention to use technology (Du & Lv, 2024; Strzelecki & ElArabawy, 2024). EE, like PE, is also a strong predictor of behavioural intentions to use technologies and studies have shown that as the ease in using a system increases, then so does the level of intention to use the technology (Tahar et al., 2020). Therefore, we propose:

Hypothesis H2: Effort expectancy (EE) is expected to have a positive influence on students' behavioural intention (BI) to use CG-AI tools for learning.

2.3.3 Social influence (SI)

SI refers to an individual's perception that those they consider important have an influence on their use of the system or technology (Venkatesh et al., 2003). In this study, SI refers to the extent to which students believe that others' opinions about CG-AI tools influences their decision to use such tools. Venkatesh et al. (2003) observed that opinions from colleagues, family, or friends can influence a user's bias to embrace new technologies. This has been supported by empirical studies showing SI to influence an individual's behavioural intention to using technologies (Du & Lv, 2024; Strzelecki & ElArabawy, 2024; Bouteraa et al., 2024). In terms of students' use of CG-AI tools for earning and research in developing countries HEI, it is expected that fellow students will be the main source of encouragement to other students in using these tools (Changalima et al., 2024). This study therefore proposes:

Hypothesis H3: Social Influence (SI) is expected to have a positive influence on students' behavioural intention (BI) to use CG-AI tools for learning.

2.3.4 Facilitating conditions (FC)

FC is an individual's perception that the level of organizational support and/or technical infrastructure can encourage the individuals use of the system (Venkatesh et al., 2003). In this study, FC is defined as students' perception that they have the technological tools and support required to use CG-AI tools for learning and research purposes. Notably, Du and Lv (2024) verified that FC is a contributing factor influencing students' use of CG-AI tools. Other studies have consistently shown the significance of FC on students' intentions to use AI tools (Strzelecki & ElArabawy, 2024). Thus, we also propose:

Hypothesis H4: Facilitating conditions (FC) is expected to have a positive influence on students' behavioural intention (BI) to use CG-AI tools for learning.

2.3.5 Attitude towards technology (ATT)

ATT is a construct adopted from the technology acceptance model (TAM) by Davis et al. (1989). It refers to an individuals' perceptions (positive or negative) about using a specific technology. TAM postulates that ATT has an influence on an individuals' behavioural intention to use technology. In the context of students' adoption of CG-AI tools, several studies have supported this relationship (Kanont et al., 2024; Ivanov et al., 2024). In this paper, ATT is defined as the positive or negative perceptions towards using CG-AI tools to aid in their research or learning process. This study therefore posits:

Hypothesis H5: Attitude Towards Technology (ATT) is expected to have a positive influence on students' behavioural intention (BI) to use CG-AI tools for learning.

2.3.6 Perceived risks (PR)

Refers to an individual's perception about the probable outcome of performing an action. This is usually due to the degree of their certainty that performing the action will result in negative consequences. As a construct, PR has been integrated with the UTAUT model to help understand why individuals use technologies such as eHealthcare IoT (Arfi et al., 2021), mobile payment apps (Widyanto et al., 2022), online shopping (Chang et al., 2016) and education (Alwahaishi, 2021; Chatterjee & Bhattacharjee, 2020). In all the studies mentioned above that have used PR as a construct, PR was observed to have a negative influence on individuals' behavioural intentions to use technology (Arfi et al., 2021; Chang et al., 2016; Chatterjee & Bhattacharjee, 2020; Widyanto et al., 2022). Similarly, in the context of using ChatGPt for assessments, PR was observed to be a demotivator to its users (Lai et al., 2024). The main risks associated with using CG-AI tools are misinformation, inaccuracies, and ethical concerns (Vázquez-Madrigal et al., 2024). In developed

countries where policies and training help to guide the use of CG-AI tools, PR is expected to have an insignificant influence on the use of such tools as the risks are reduced (Chatterjee & Bhattacharjee, 2020). However, in developing countries where these tools are still yet to be understood, it is expected that:

Hypothesis H6: Perceived Risks (PR) is expected to have a negative influence on students' behavioural intention (BI) to use CG-AI tools for learning.

3 Methodology

This section outlines the research approach taken to investigate students' perception of using CG-AI tools for their learning and research processes. Details on the research design, participants, data collection methods, and data analysis techniques are presented here. The research design is focused on addressing the knowledge gap identified in the literature review section. The purpose of the study is to determine the factors responsible for Nigerian students' use of CG-AI tools for learning and research purposes.

Based on the population characteristics and the objectives of the study, purposive sampling method was used to recruit the study participants and a quantitative methodology was adopted like previous studies founded on technology adoption models. To empirically validate the hypothesis formulated in this study, data was collected using a paper-based survey and the target population were all registered students, in the Department of Computer Science, at a state university in northern Nigeria. This included both undergraduate and postgraduate (MSc and PhD) students. There were no restrictions regarding the target population in relation to age or gender however, this information was collected to aid in describing the respondents.

304 responses were received from the participants and after eliminating surveys that were incomplete, a total of 289 valid responses were obtained. The survey was administered in the national language (English). Prior to the collection of data, 2 reviewers assisted in reviewing the survey to establish the relevance of the questions to the research objectives, and the overall structure of the survey. The reviewers included an associate professor in the field of Information Systems (PhD. Information Systems) and a professor in the domain of Computer Science (PhD. Computer Science). Both reviewers have a combination of over 36 years of experience in teaching and research of computing subjects and a combined total of over 122 published journal articles and conference papers.

The survey instrument was made up of 2 sections. Section A collected demographic information such as age, gender, level of study and the AI tool of choice for learning or research purposes. Age was grouped into categories (Below 20, 21 – 30, 31 – 40, 41 – 50, and Over 50). Dummy variables were used to code gender as 1 and 2 to represent male and female, respectively. Level of study was captured as either Undergraduate or Postgraduate. The variables Yes or No were required to answer the question pertaining to support in using the CG-AI tool from either the lecturer or the school. Finally, the question relating to the choice of the CG-AI tool used by

the respondents was open-ended to allow them to respond with the name of their preferred tool.

Section B contained Likert-styled statements to collect the students' responses in relation to the constructs used in the research model (Performance Expectancy, Social Influence, Effort Expectancy, Facilitating Conditions, Attitude Towards Technology, Perceived Risks, and Behavioural Intention). The constructs were measured with 4 or 5 items using a 5—point Likert scale which ranged from strongly agree (1) to strongly disagree (5).

Prior to administering the survey, a validation of the survey instrument was done via a pilot test where 20 students were administered the survey to ensure that survey wordings were unambiguous, and the questions were reliable and valid. Consent was sought from the respondents to notify them of the benefits and risks associated with the research and to assure them that their responses will be completely anonymous.

To determine the factors responsible for influencing students use of CG-AI tools' the proposed research model was used and the constructs were tested using structural equation modelling (SEM). IBM's Statistical Package for the Social Sciences (SPSS) and Analysis of Moment Structures (AMOS) were the chosen tools to analyse the data and to test all 6 hypotheses. A two-step approach was used to investigate the relationship between the constructs (Anderson & Gerbing, 1988).

4 Data analysis and results

As mentioned in the previous section, the collected data was analysed using SPSS. To understand the respondents' demography, descriptive analysis was performed. Table 1 shows that 70.20% of the respondents were male while 29.8% were female. In terms of the age groups, over half of the participants (50.20%) were between the ages of 21 – 30 years. This was followed by 28% for the ages 31 – 40 years. 14.20% were below the ages of 20 years, 6.90% between the ages of 41 – 50 years, and 0.70% over 50 years of age. Regarding the level of studies, 66.80% identified

Table 1 Respondents demographic data ($n = 289$)

Demographic Characteristics		Frequency	Percentage (%)
Gender	Male	203	70.20
	Female	86	29.80
Age	Below 20	41	14.20
	21—30	145	50.20
	31—40	81	28.00
	41—50	20	6.90
	Over 50	2	0.70
Level of Study	Undergraduate	193	66.80
	Postgraduate	96	33.20
Support from Lecturer or School	Yes	96	22.10
	No	193	77.90

as undergraduates and 33.20% were at the postgraduate level. Finally, only 22.10% claimed that they received support from their lecturers/school in using CG-AI tools for learning and research purposes as opposed to 77.90% who did not.

Table 2 below shows the choice of AI tools as informed by the participants. ChatGPT seems to be the preferred AI tool with almost half the responses (47.30%), closely followed by Gemini with 29.50%. The least used tool is Deep Chat (0.40%) followed by Elicit (1.10%).

In the second section of the survey instrument, where comments were used to collect responses based on the constructs in the hypothetical model, structural equation modelling (SEM) was used to analyse the data. Firstly, multicollinearity tests were conducted on the 6 constructs data based on the dependent variable (BI). The results, Table 3, showed that the Variance Inflation Factor (VIF) and tolerance values fall within the accepted values. VIF and Tolerance are measures of the degree of multicollinearity (O'brien, 2007). A VIF of over 4.00 and tolerance of less than 0.25 indicate potential multicollinearity problems (Hair et al., 2010; Mason et al., 1989). The resulting VIF range of 1.41 to 3.04 is less than 4 while the tolerance values are all above 0.25 indicating inconsequential collinearity.

4.1 Measurement model

The descriptive analysis for the constructs used in the research model are captured in Table 4. The means for all constructs, except for ATT and PR, are all under 2

Table 2 AI usage of tools ($n = 264$)

AI Tool	Frequency	Percentage (%)
AI Chat	11	4.20
Ask QX	7	2.70
Chat GPT	125	47.30
Gemini/ Bard	78	29.50
Deep Chat	1	0.40
Elicit	3	1.10
Aithor	39	14.80

Table 3 Multicollinearity results

Dependent variable: Behavioral Intention (BI)		
Construct	*Tolerance*	*VIF*
SI	0.34	3.04
EE	0.49	2.20
PE	0.72	1.41
PR	0.61	1.85
ATT	0.46	2.18
FC	0.40	2.57

Table 4 Descriptive analysis of the constructs

Construct	Mean	Std Deviation	Cronbach's Alpha	Number of items
FC	1.749	0.716	0.892	4
ATT	2.931	1.107	0.873	4
PR	3.318	0.959	0.718	4
BI	1.544	0.663	0.871	4
PE	1.567	0.504	0.718	4
EE	1.688	0.644	0.843	4
SI	1.590	0.708	0.906	4

showing that the responses for the attributes are positive. Internal reliability was established via Cronbach's alpha, George and Mallery (2003) suggests that Cronbach's alpha values are good if they are over 0.8 and excellent if over 0.9. All the constructs displayed values of over 0.8 except for PR (0.718) this however is acceptable according to Salkind (2015).

To understand the relationships that exist between the constructs within the research model, a two-step approach was employed. Firstly, a confirmatory factor analysis based on the maximum-likelihood estimation (MLE) method was used to estimate model's parameters. MLE is a method used to determine parameters such as means and standard deviations, of a normally distributed random sample data (Wasserman, 2013).

The next step was to conduct a goodness-of-fit test based on the proposed model. Table 4 captures the actual measurement model values in comparison to the recommended values for the fit indicators as recommended by Kline (2015), Hu and Bentler (1999), and Hair and et al., (2010). All measured indices were observed to fall within the suggested values. To confirm the reliability of the measured factors, composite reliability (CR) was used and as captured in in Table 5, all items measured exhibited a CR reading above the recommended value of 0.70 (Hair et al., 2010). Validity tests were run on the constructs to measure the average variance extracted (AVE). The AVE of all constructs was observed to be above the recommended value of 0.5 and this was also measured to be less than the equivalent CR. Discriminant validity was established by ensuring that the ASV (average shared

Table 5 Construct reliability, convergent validity, discriminant validity, and factor correlation matrix

	CR	AVE	MSV	MaxR(H)	PE	SI	FC	ATT	BI	EE	PR
PE	0.865	0.618	0.218	0.742	**0.794**						
SI	0.852	0.658	0.303	0.857	0.446	**0.811**					
FC	0.895	0.682	0.019	0.911	0.091	−0.017	**0.792**				
ATT	0.873	0.633	0.048	0.877	0.198	0.219	0.139	**0.712**			
BI	0.873	0.632	0.262	0.879	0.467	0.512	0.112	0.067	**0.752**		
EE	0.852	0.595	0.303	0.893	0.164	0.550	0.008	0.107	0.090	**0.882**	
PR	0.844	0.645	0.017	0.873	0.130	0.103	0.107	0.121	0.069	0.062	**0.834**

Table 6 Model fit indices table

Fit Index	Recommended Value	Measurement Model	Structural Model
CHI Square	N/A	537.867	537.867
Degree of Freedom (df)	N/A	274	274
P Value	P-value > 0.050	0.000	0.000
CFI	> 0.900	0.940	0.940
GFI	> 0.900	0.922	0.918
AGFI	> 0.800	0.803	0.803
RMSEA	< 0.080	0.059	0.068
RMSR	< 0.100	0.052	0.054
NFI	> 0.900	0.911	0.911

CFI comparative fit index, *GFI* goodness-of-fit index, *AGFI* adjusted goodness-of-fit, *RMSEA* root mean square error of approximation, *RMSR* root mean square residuals, *NFI* normed fit index

Table 7 Structural model (Significance level at ***$p < 0.001$)

Hypothesis	Path	Estimate	P-value	Outcome
H1	PE –> BI	0.551	***	Supported
H2	EE –> BI	0.466	***	Supported
H3	SI–> BI	0.507	***	Supported
H4	FC –> BI	0.003	0.963	Unsupported
H5	PR –> BI	−0.322	0.585	Unsupported
H6	ATT –> BI	0.031	0.991	Unsupported

variance) and the MSV (maximum shared variance) were less than the corresponding AVE value (Hair et al., 2010). Also, the square root of the AVE is expected to be higher than the correlation value (Bagozzi & Yi, 1988). Based on these conditions, discriminant validity was established. The next step was to test hypothesis by investigating the research model (structural measurement) considering all constructs have shown to be reliable and valid.

4.2 Structural model

To examine the relationships between the constructs, a structural model was built. Table 6 shows the goodness-of-fit readings for the structural model are all within the recommended values indicating that the data fits the structural model.

Based on the research model, Table 7 below shows the association between the hypothesized variables. It captures the path coefficients indicating that all the hypothesized relationships were supported except for hypothesis H4, H5, and H6. PE ($\alpha = 0.551$, $p < 0.001$), EE ($\alpha = 0.466$, $p < 0.001$) and SI ($\alpha = 0.507$, $p < 0.001$) had a positive and significant effect on BI. PR had a negative but insignificant influence on BI while FC and ATT were observed to have an insignificant effect on BI. Thus,

hypotheses H1, H2, and H3 were supported but there was no support for hypotheses H4, H5, and H6.

5 Discussion

As established earlier, the purpose of this study is to identify the factors that influence students' behavioural intentions to use CG-AI tools for learning and research purposes in the context of a developing country and specifically in northern Nigeria. The research utilized an adapted version of the UTAUT model. The findings of this study provide valuable insights into how CG-AI tools are perceived by students and the factors that influence the adoption of such tools.

Based on the disruptive effect of CG-AI tools, there has been a rise in issues pertaining academic integrity and plagiarism. Therefore, there is a need to address CG-AI usage in HEIs (Moorhouse et al., 2023). Students need to be informed of the ethical ways to use CG-AI tools as well as the penalties that will be applied for its misuse. Educational institutions also need to develop policies and guidelines on its use (Moorhouse et al., 2023). From the descriptive statistics in Table 1, most students (77.90%) who participated in the study declared that they did not receive any guidance from their lecturers or institutions in using CG-AI tools for learning, research, or assessments. This implies that learning is not supported and if there are no checks on assessment submissions academic integrity issues will be prevalent.

Students identified ChatGPT (47.30%) as their preferred CG-AI tool as captured in Table 2, this was followed by Google's Gemini (29.50%). While these tools all generate content, the quality and style differ based on the training data set used (Bandi et al., 2023). This means that a query can have different responses based on the CG-AI tool used. Some tools might be best suited for research, while others will be better at providing generic information. Further investigation would be required to understand why these tools are the preferred choice of students for research and learning purposes.

Aligning with previous studies, the results of this study indicate that PE is a primary determinant of students' behavioural intentions to use CG-AI tools for learning and research purposes. This has been corroborated by similar studies related to learning tools such as ChatGPT (Strzelecki, 2024), mobile learning systems (Almaiah et al., 2019), and learning management systems (Al-Adwan et al., 2022). The results emphasize the benefits of CG-AI tools in relation to student's performance in their academic pursuits which in-turn encourage their intention to use such tools. The rationale behind this could be attributed to the time saving nature of the tools to complete learning and research activities (Strzelecki & ElArabawy, 2024) and the ability to generate quality output in response to a user's prompt (Chan & Hu, 2023).

With regards to H2, the relationship between EE and BI was observed to be positive and significant aligning with previous studies related students' perceptions of learning tools such as learning management systems (Al-Mamary, 2022), and ChatGPT (Strzelecki, 2024; Strzelecki & ElArabawy, 2024). This means that students

are inclined to use CG-AI tools as they find it easy to use requiring very little effort. This result must be interpreted with caution, as this study did not investigate how students use the tools. To effectively and ethically use CG-AI tools, effort and time is required to verify the generated content as well as rewriting it in the student's voice.

The final supported hypothesis was observed in the relationship between SI and BI. This finding was corroborated by studies that investigated students behavioural intention to use learning tools such as mobile learning (Alowayr, 2022; Kang et al., 2015) and e-learning systems (El-Masri & Tarhini, 2017). However other studies have differing results, for example in the context of Egyptian students' use of ChatGPT (Strzelecki & ElArabawy, 2024), and Nigerian students with learning managemnt systems (Yakubu & Dasuki, 2019). The findings suggest that the use of CG-AI tools by students are influenced by significant figures especially their peers as most of the students reported that they did not receive support/guidance from the school administration or from their instructors. However, as Nigerian HEIs start to develop guidelines for CG-AI tools the significance of SI is expected to increase.

The study revealed that FC is not a predictor of students' behavioural intentions to use CG-AI tools, thus hypothesis H4 was not supported. This does not echo the findings of previous studies related to learning technologies such as learning management systems (Al-Adwan et al., 2022), mobile learning systems (Almaiah et al., 2019) and ChatGPT (Strzelecki & ElArabawy, 2024). However, it agrees with other research studies on the acceptance MOOCs (Altalhi, 2021) and ChatGPT (Faruk et al., 2023; Strzelecki, 2024). The results indicate that students do not believe that access to organizational and technological infrastructure are available to facilitate the use of CG-AI tools. This can be explained, to an extent, by the demographic data showing that most students (77.9%) did not get support in using the tools by their lecturer or school. Another reason could be that students only require access to an internet ready device thus there is no other support needed nor access to complex technologies to access CG-AI tools. Finally, CG-AI tools are easy to use and similar in functionality and "look and feel" of popular search engines like Edge or Google, therefore, additional knowledge and support is no longer required (Faruk et al., 2023).

In the relationship between ATT and BI, analysis of the collected data shows that there is no influence on students' attitude towards CG-AI tools and their behavioural intentions to use them. This means that the learners' positive perceptions about have no effect on their use of the tools. This result to contrary to findings from similar adoption studies in AI in content creation (Anandaputra et al., 2024) and ChatGPT (Saif et al., 2024). We attribute our findings to the fact that more than half the respondents (64.40%) fall within the Gen Z age group (under 30), and they are more likely to use novel technologies, even if they have negative attitudes towards using technologies in general. Another factor to consider is that the student's might be under pressure to submit and pass their courses/modules which simulates a mandatory setting where the students must use the tools regardless of their attitudes.

Finally, hypothesis H6 (the relationship between PR and BI) was not supported. Even though a negative influence was observed the results were not significant. A significant and negative effect on BI was expected from PR as we believe that if students are aware of the penalties applied to the use of CG-AI tools incorrectly, it would deter them from using it. This is especially true considering that there might be concerns pertaining to accuracy and factuality the responses received from GG-AI tools. Similar studies corroborate our findings (Chatterjee & Bhattacharjee, 2020). This study attributes the insignificant results to the fact that students are aware that their assessment submissions will not be checked for AI misconduct. Also, it is envisaged that students use these tools mainly to generate learning content to aid them with their studies. Further research on how Nigerian students use CG-AI tools for learning and research will help better understand why this relationship was not supported.

The findings of this study can help formulate policies and guidelines that support the use of CG-AI tools in higher education, specifically in developing countries such as Nigeria. Considering that PE was observed to influence students' behavioural intentions to use these tools, it implies that students believe that their performance in learning will be improved by using CG-AI tools. Their use of these tools is further supported by, social influence, i.e., the influence of those they consider are important to them. However, the risks involved in using CG-AI for learning might not be fully appreciated. Therefore, policy makers can develop existing academic integrity policies which capture acceptable use of technologies for learning and assessments. Prevention and detection strategies can be employed to deter academic misconduct. For example, alternative assessments can be modified to focus on and support higher order thinking skills. Finally, institutional ethical frameworks can be used to address issues such as bias, fairness, transparency, and accountability when using AI for learning, research, and assessments.

6 Conclusion

The widespread use of CG-AI tools like ChatGPT and Google's Gemini has sparked interests surrounding students' perspectives on these tools for learning and research. Prior research suggests that students' learning outcomes are influenced by their approach to learning (Biggs, 1999). Their learning approach results from their perceptions of the learning environments or tools they are exposed to, the individual's abilities, and the strategies used to teach them (Chan & Hu, 2023). With regards to the learning environment/tools, this study aimed at investigating the factors that influence Nigerian students' behavioural intentions to use CG-AI tools. The study adopted the UTAUT framework, incorporating "*perceived risks*" and "*attitude towards technology*" constructs. A total of 6 hypotheses were tested of which 3 were supported. PE, SI and EE were all identified as determinants of students' behavioural intentions to use CG-AI tools. There was no support for ATT, FC, and PR.

Structural equation modelling (SEM) was used to analyse and model the data collected to empirically validate the significance of Nigerian students' views of

using CG-AI tools for learning and research purposes. The main benefactors of the results are developers, researchers, and academic administrators. Developers can use the findings to enhance CG-AI tools. Academic administrators can develop policies and strategies for integrating CG-AI tools into education. Researchers can further investigate additional factors hypothesized to influence students use of CG-AI tools.

6.1 Research and practice implications

There are significant practical and theoretical implications derived from this research relating to the use of CG-AI tools by Nigerian students. The UTAUT model is a comprehensive model that helps in understanding the factors that influence an individual to use technology, however, there are very few studies that validate the model in the context of students' behavioural intentions to use CG-AI tools, specifically in a developing country where policies and guidelines are still being developed. The study therefore contributes to content generation tools literature by assessing the UTAUT framework in the context of a developing country especially when considering the cultural and societal differences in comparison to developed countries. In terms of practical implications, the constructs used in this research can help guide HEI administrators on the factors that influence students use of CG-AI tools. For example, considering FC was not a determinant of behavioural intention, university administrators can help to guide and support students in using CG-AI tools effectively so that learning takes place and academic misconduct is avoided.

6.2 Limitations and future research

This research was conducted on students, within a department, and selected from a single HEI in northern Nigeria. Therefore, the results may not adequately capture students' perceptions towards the use of CG-AI tools for the entire student population of Nigeria, let alone other developing countries. Future research could be conducted based on a wider scope of students and take into consideration moderating factors such as gender, age, and experience.

This research focused on CG-AI tools in general, this could significantly influence the findings. Research stemming from this study could focus on a specific tool that has been adopted by the institution where training and support has been provided.

Quantitative research is subject to generalised; hence, future research could adopt a qualitative or mixed-methods approaches to provide deeper understanding on the determinants of students' behavioural intentions to use CG-AI tools.

Finally, alternative insights could be revealed by testing other theoretical frameworks for example, the DeLone and McLean IS Success model (DeLone & McLean, 1992), or by adding unique constructs such as habit (Alhwaiti, 2023) and perceived ethics (Gajić et al., 2024).

Appendix: Survey instruments

Section A: Student's Demographics

Age (years)
- ○ Below 20
- ○ 21 – 30
- ○ 31 – 40
- ○ 41 – 50
- ○ Over 50

What is your gender?
- ○ Male
- ○ Female
- ○ Prefer not to say.

What is your level of study?
- ○ Undergraduate
- ○ Postgraduate

Does the University or your lecturer support or guide you in using AI for your assessments?
- ○ Yes
- ○ No

Which AI tool (if any) do you use for learning or research (e.g., Chat GPT, Elicit, etc.)

_____

Section B: Student's Perceptions on the use of AI tools for Learning

Performance Expectancy	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
PE1: I find AI technologies useful in my studies.					
PE2: Using AI technologies enables me to accomplish tasks more quickly.					
PE3: Using AI technologies increases my productivity.					
PE4: If I use AI technologies, I will increase my chances of getting a better grade.					

Effort Expectancy	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
EE1: My interaction with AI technologies is clear and understandable					
EE2: It is easy for me to become skillful at using AI technologies.					
EE3: I find using AI technologies easy to use.					
EE4: Learning to operate AI technologies is easy for me.					

Social Influence	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
SI1: My instructors encourage me to use AI technologies for learning and research					
SI2: My peers (classmates) encourage me to use AI technologies for learning and research					
SI3: The University (management/administration) encourage me to use AI technologies for learning and research					
SI4: Generally speaking, People who are important to me think that I should use AI technologies for learning and research					

Facilitating Conditions	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
FC1: I have all the necessary resources to					

use AI technology for teaching and learning					
FC2: I have the knowledge necessary to use AI for learning and research					
FC3: My lecturer/school are available to assist me with using AI for learning and research					
FC4: My school has all the necessary resources to use AI technology for teaching learning and research.					

Attitude towards technology	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
ATT1: Using AI technologies for learning and research is a good idea.					
ATT2: Using AI technologies makes learning and research interesting.					
ATT3: Working with AI technologies makes learning and research fun.					
ATT4: I like working with AI technologies for learning and research					

Perceived Risk	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
PR1: I believe AI-powered educational content is not always correct					
PR2: My lecturers and the school do not support my use of AI technology for learning and research					
PR3: The use of AI for learning and research purpose is confusing					
PR4: Use of AI technology for learning and research purpose is risky					

Behavioral Intentions	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
BI1: I intend to use AI technologies for learning and research in the near future					
BI2: I predict I will use AI technologies for learning and research in the near future					
BI3: I plan to use AI technologies frequently for learning and research					
BI4: When given the chance, I will always try to use AI technologies for learning and research.					

Data availability The data set generated and analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflicts of interest The authors have no relevant financial or non-financial interests to disclose.
The authors have no conflicts of interest to declare that are relevant to the content of this article.
All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.
The authors have no financial or proprietary interests in any material discussed in this article.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Abbad, M. M. (2021). Using the UTAUT model to understand students' usage of e-learning systems in developing countries. *Education and Information Technologies, 26*(6), 7205–7224. https://doi.org/10.1007/s10639-021-10573-5

Adarkwah, M. A., Amponsah, S., van Wyk, M. M., Huang, R., Tlili, A., Shehata, B., Metwally, A. H. S., & Wang, H. (2023). Awareness and acceptance of ChatGPT as a generative conversational AI for transforming education by Ghanaian academics: A two-phase study. *Journal of Applied Learning and Teaching, 6*(2), 1–6. https://doi.org/10.37074/jalt.2023.6.2.26

Al-Adwan, A. S., Yaseen, H., Alsoud, A., Abousweilem, F., & Al-Rahmi, W. M. (2022). Novel extension of the UTAUT model to understand continued usage intention of learning management systems: The role of learning tradition. *Education and Information Technologies, 27*(3), 3567–3593. https://doi.org/10.1007/s10639-021-10758-y

Alhwaiti, M. (2023). Acceptance of artificial intelligence application in the post-covid era and its impact on faculty members' occupational well-being and teaching self-efficacy: A path analysis using the UTAUT 2 model. *Applied Artificial Intelligence, 37*(1), 2175110. https://doi.org/10.1080/08839514.2023.2175110

Almaiah, M. A., Alamri, M. M., & Al-Rahmi, W. (2019). Applying the UTAUT model to explain the students' acceptance of mobile learning system in higher education. *IEEE Access, 7*, 174673–174686.

Al-Mamary, Y. H. (2022). Understanding the use of learning management systems by undergraduate university students using the UTAUT model: Credible evidence from Saudi Arabia. *International Journal of Information Management Data Insights, 2*(2), 100092. https://doi.org/10.1016/j.jjimei.2022.100092

Alowayr, A. (2022). Determinants of mobile learning adoption: Extending the unified theory of acceptance and use of technology (UTAUT). *International Journal of Information and Learning Technology, 39*(1), 1–12. https://doi.org/10.1108/IJILT-05-2021-0070

Altalhi, M. (2021). Toward a model for acceptance of MOOCs in higher education: The modified UTAUT model for Saudi Arabia. *Education and Information Technologies, 26*(2), 1589–1605. https://doi.org/10.1007/s10639-020-10317-x

Al-Tkhayneh, K. M., Alghazo, E. M., & Tahat, D. (2023). The advantages and disadvantages of using artificial intelligence in Education. *Journal of Educational and Social Research, 13*(4), 105–117. https://doi.org/10.36941/jesr-2023-0094

Alwahaishi, S. (2021). Student use of e-learning during the coronavirus pandemic: An extension of UTAUT to trust and perceived risk. *International Journal of Distance Education Technologies (IJDET), 19*(4), 72–90.

Anandaputra, G. V., Yungistira, G. A., Nicholas, O., & Mailangkay, A. B. (2024). Analysis on User's Attitude Towards Content Creation Using Artificial Intelligence in Social Media. *3rd International Conference on Creative Communication and Innovative Technology (ICCIT),* (pp. 1–6). https://doi.org/10.1109/ICCIT62134.2024.10701251

Anderson, J. C., & Gerbing, D. W. (1988). Structural equation modeling in practice: A review and recommended two-step approach. *Psychological Bulletin, 10*(3), 411.

Andrews, J. E., Ward, H., & Yoon, J. (2021). UTAUT as a model for understanding intention to adopt AI and related technologies among librarians. *The Journal of Academic Librarianship, 47*(6), 102437. https://doi.org/10.1016/j.acalib.2021.102437

Arenas-Gaitán, J., Ramírez-Correa, P. E., & Rondán-Cataluña, F. J. (2011). Cross cultural analysis of the use and perceptions of web based learning systems. *Computers & Education, 57*(2), 1762–1774. https://doi.org/10.1016/j.compedu.2011.03.016

Arfi, W. B., Nasr, I. B., Khvatova, T., & Zaied, Y. (2021). Understanding acceptance of eHealthcare by IoT natives and IoT immigrants: An integrated model of UTAUT, perceived risk, and financial cost. *Technological Forecasting and Social Change, 163*, 12. https://doi.org/10.1016/j.techfore.2020.120437

Bagozzi, R. P., & Yi, Y. (1988). On the evaluation of structural equation models. *Journal of the Academy of Marketing Science, 16*(1), 74–94. https://doi.org/10.1007/BF02723327

Baidoo-Anu, D., & Ansah, L. O. (2023). Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. *Journal of AI, 7*(1), 52–62. https://doi.org/10.61969/jai.1337500

Bandi, A., Adapa, P. V., & Kuchi, Y. E. (2023). The power of generative ai: A review of requirements, models, input–output formats, evaluation metrics, and challenges. *Future Internet, 15*(8), 260. https://doi.org/10.3390/fi15080260

Biggs, J. (1999). What the student does: Teaching for enhanced learning. *Higher Education Research & Development, 18*(1), 57–75. https://doi.org/10.1080/0729436990180105

Bouteraa, M., Bin-Nashwan, S. A., Al-Daihani, M., Dirie, K. A., Benlahcene, A., Sadallah, M., Zaki, H. O., Lada, S., Ansar, R., Fook, L. M., & Chekima, B. (2024). Understanding the diffusion of AI-generative (ChatGPT) in higher education: Does students' integrity matter? *Computers in Human Behavior Reports, 14*, 100402. https://doi.org/10.1016/j.chbr.2024.100402

Chan, C. Y. (2023). A comprehensive AI policy education framework for university teaching and learning. *International Journal of Educational Technology in Higher Education, 20*(1), 38. https://doi.org/10.1186/s41239-023-00408-3

Chan, C. K., & Hu, W. (2023). Students' voices on generative AI: Perceptions, benefits, and challenges in higher education. *International Journal of Educational Technology in Higher Education, 20*(1), 43. https://doi.org/10.1186/s41239-023-00411-8

Chang, H. H., Fu, C. S., & Jain, H. T. (2016). Modifying UTAUT and innovation diffusion theory to reveal online shopping behavior: Familiarity and perceived risk as mediators. *Information Development, 32*(5), 1757–1773. https://doi.org/10.1177/0266666915623317

Changalima, I. A., Amani, D., & Ismail, I. J. (2024). Social influence and information quality on Generative AI use among business students. *The International Journal of Management Education, 22*(3), 101063. https://doi.org/10.1016/j.ijme.2024.101063

Chatterjee, S., & Bhattacharjee, K. K. (2020). Adoption of artificial intelligence in higher education: A quantitative analysis using structural equation modelling. *Education and Information Technologies, 25*, 3443–3463. https://doi.org/10.1007/s10639-020-10159-7

Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1989). *Technology Acceptance Model. J Manag Sci, 35*(8), 982–1003. https://doi.org/10.1007/978-3-030-45274-2

DeLone, W. H., & McLean, E. R. (1992). Information systems success: The quest for the dependent variable. *Information Systems Research., 3*(1), 60–95. https://doi.org/10.1287/isre.3.1.60

Denny, P., Khosravi, H., Hellas, A., Leinonen, J., & Sarsa, S. (2023). Can we trust AI-generated educational content? Comparative analysis of human and AI-generated learning resources. *arXiv:*2306.10509.

Du. , L., & Lv, B. (2024). Factors influencing students' acceptance and use generative artificial intelligence in elementary education: an expansion of the UTAUT model. *Education and Information Technologies*, 1–20. https://doi.org/10.1007/s10639-024-12835-4

Edo, O. C., Ang, D., Etu, E. E., Tenebe, I., Edo, S., & Diekola, O. A. (2023). Why do healthcare workers adopt digital health technologies-A cross-sectional study integrating the TAM and UTAUT model in a developing economy. *International Journal of Information Management Data Insights, 3*(2), 100186. https://doi.org/10.1016/j.jjimei.2023.100186

El-Masri, M., & Tarhini, A. (2017). Factors afecting the adoption of e-learning systems in Qatar and USA: Extending the unifed theory of acceptance and use of technology 2 (UTAUT2). *Educational Technology Research and Development, 65*(3), 743–763. https://doi.org/10.1007/s11423-016-9508-8

Ernst, C. P. (2017). The influence of disgust on technology acceptance. *Twenty-third Americas Conference on Information Systems (AMCIS 2017 Proceedings),* Boston, pp. 539–547.

Essien, A., Salami, A., Ajala, O., Adebisi, B., Shodiya, A., & Essien, G. (2024). Exploring socio-cultural influences on generative AI engagement in Nigerian higher education: An activity theory analysis. *Smart Learning Environments, 11*(1), 63.

Faruk, L. D., Rohan, R., Ninrutsirikun, U., & Pal, D. (2023). University Students' Acceptance and Usage of Generative AI (ChatGPT) from a Psycho-Technical Perspective. *13th International Conference on Advances in Information Technology*, 1–8. https://doi.org/10.1145/3628454.3629552

Feuerriegel, S., Hartmann, J., Janiesch, C., & Zschech, P. (2024). Generative AI. *Business & Information Systems Engineering, 66*(1), 111–126.

Gajić, T., Vukolić, D., Bugarčić, J., Đoković, F., Spasojević, A., Knežević, S., ĐorđevićBoljanović, J., Glišić, S., Matović, S., & Dávid, L. D. (2024). The Adoption of Artificial Intelligence in Serbian Hospitality: A Potential Path to Sustainable Practice. *Sustainability, 16*(8), 3172. https://doi.org/10.3390/su16083172

George, D., & Mallery, P. (2003). SPSS for Windows step by step: answers to selected exercises. *A simple guide and reference, 63*(1), 1461–1470.

Hadi, M. U., Al Tashi, Q., Shah, A., Qureshi, R., Muneer, A., Irfan, M., Zafar, A., Shaikh, M. B., Akhtar, N., Wu, J., & Mirjalili, S. (2024). Large language models: A comprehensive survey of its applications, challenges, limitations, and future prospects. *Authorea Preprints*. https://doi.org/10.36227/techrxiv.23589741.v6

Hair, J. F., Anderson, R. E., Tatham, R. L., & Black, W. C. (2010). *Multivariate Data Analysis*. Prentice Hall.

Hanafizadeh, P., Behboudi, M., Koshksaray, A. A., & Tabar, M. J. S. (2014). Mobile-banking adoption by Iranian bank clients. *Telematics and Informatics, 31*(1), 62–78. https://doi.org/10.1016/j.tele.2012.11.001

Hu, K., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modelling: A Multidisciplinary Journal, 6*(1), 1–55. https://doi.org/10.1080/10705519909540118

Hutson, J. (2024). Rethinking Plagiarism in the Era of Generative AI. *Journal of Intelligent Communication, 4*(1), 20–31.

Ivanov, S., Soliman, M., Tuomi, A., Alkathiri, N. A., & Al-Alawi, A. N. (2024). Drivers of generative AI adoption in higher education through the lens of the Theory of Planned Behaviour. *Technology in Society, 77*, 102521. https://doi.org/10.1016/j.techsoc.2024.102521

Kang, M., Liew, B. Y., Lim, H., Jang, J., & Lee, S. (2015). Investigating the determinants of mobile learning acceptance in korea using UTAUT2. In: Chen, G., Kumar, V., Kinshuk, Huang, R. & Kong, S. C. (eds.), *Lecture Notes in Educational Technology*, pp. 209–216. https://doi.org/10.1007/978-3-662-44188-6_29

Kanont, K., Pingmuang, P., Simasathien, T., Wisnuwong, S., Wiwatsiripong, B., Poonpirome, K., Songkram, N., & Khlaisang, J. (2024). Generative-AI, a Learning Assistant? Factors Influencing Higher-Ed Students' Technology Acceptance. *Electronic Journal of e-Learning, 22*(6), 18–33. https://doi.org/10.34190/ejel.22.6.3196

Kasneci, E., Seßler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., Gasser, U., Groh, G., Günnemann, S., Hüllermeier, E., & Krusche, S. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences, 103*, 102274. https://doi.org/10.1016/j.lindif.2023.102274

Kline, R. B. (2015). *Principles and Practice of Structural Equation Modeling*. Guilford publications.

Krause, S., Panchal, B. H., & Ubhe , N. (2024). The Evolution of Learning: Assessing the Transformative Impact of Generative AI on Higher Education. *arXiv preprint arXiv:*2404.10551. https://doi.org/10.48550/arXiv.2404.10551

Kurdi, G., Leo, J., Parsia, B., Sattler, U., & Al-Emari, S. (2020). A systematic review of automatic question generation for educational purposes. *International Journal of Artificial Intelligence in Education, 30*, 121–204. https://doi.org/10.1007/s40593-019-00186-y

Lai, C. Y., Cheung, K. Y., Chan, C. S., & Law, K. K. (2024). Integrating the adapted UTAUT model with moral obligation, trust, and perceived risk to predict ChatGPT adoption for assessment support: A survey with students. *Computers and Education: Artificial Intelligence, 6*, 100246. https://doi.org/10.1016/j.caeai.2024.100246

Lin, X. (2024). Exploring the role of ChatGPT as a facilitator for motivating self-directed learning among adult learners. *Adult Learning, 35*(3), 156–166. https://doi.org/10.1177/10451595231184928

Mason, R. L., Gunst, R. F., & Hess, J. L. (1989). Statistical design and analysis of experiments: Applications to engineering and science. Wiley.

Moorhouse, B. L., Yeo, M. A., & Wan, Y. (2023). Generative AI tools and assessment: Guidelines of the world's top-ranking universities. *Computers and Education Open, 5*, 100151. https://doi.org/10.1016/j.caeo.2023.100151

Murugesan, S., & Cherukuri, A. K. (2023). The rise of generative artificial intelligence and its impact on education: The promises and perils. *Computer, 56*(5), 116–121. https://doi.org/10.1109/MC.2023.3253292

Ob'rien, R. M. (2007). A caution regarding rules of thumb for variance inflation factors. *Quality and Quantity, 41*(5), 673–690. https://doi.org/10.1007/s11135-006-9018-6

Rachmawati, I. K., Bukhori, M., Majidah, Y., & Hidayatullah, S. (2020). Analysis of use of mobile banking with acceptance and use of technology (UTAUT). *International Journal of Scientific and Technology Research, 9*(8), 534–540.

Saif, N., Khan, S. U., Shaheen, I., Alotaibi, F. A., Alnfiai, M. M., & Arif, M. (2024). Chat-GPT; validating Technology Acceptance Model (TAM) in education sector via ubiquitous learning mechanism. *Computers in Human Behavior, 154*, 108097. https://doi.org/10.1016/j.chb.2023.108097

Salkind, N. (2015). *Encyclopedia of measurement and statistics* (1st ed.). Sage.

Sharma, S., Islam, N., Singh, G., & Dhir, A. (2022). Why do retail customers adopt artificial intelligence (AI) based autonomous decision-making systems? *IEEE Transactions on Engineering Management, 71*, 1846–1861. https://doi.org/10.1109/TEM.2022.3157976

Strzelecki, A. (2024). Students' acceptance of ChatGPT in higher education: An extended unified theory of acceptance and use of technology. *Innovative Higher Education, 49*(2), 223–245. https://doi.org/10.1007/s10755-023-09686-1

Strzelecki, A., & ElArabawy, S. (2024). Investigation of the moderation effect of gender and study level on the acceptance and use of generative AI by higher education students: Comparative evidence from Poland and Egypt. *British Journal of Educational Technology, 55*(3), 1209–1230. https://doi.org/10.1111/bjet.13425

Šumak, B., Heričko, M., Polančič, G., & Pušnik, M. (2010). Investigation of e-learning system acceptance using UTAUT. *International Journal of Engineering Education, 26*(6), 1327.

Tahar, A., Riyadh, H. A., Sofyani, H., & Purnomo, W. E. (2020). Perceived ease of use, perceived usefulness, perceived security, and intention to use e-filing: The role of technology readiness. *The Journal of Asian Finance, Economics and Business, 7*(9), 5. https://doi.org/10.13106/jafeb.2020.vol7.no9.537

Tiwari, C. K., Bhat, M. A., Khan, S. T., Subramani, R., & Khan, M. A. (2024). What drives students toward ChatGPT? An investigation of the factors influencing adoption and usage of ChatGPT. *Interactive Technology and Smart Education, 21*(3), 333–355. https://doi.org/10.1108/ITSE-04-2023-0061

Vázquez-Madrigal, C., García-Rubio, N., & Triguero, Á. (2024). Generative Artificial Intelligence in Education: Risks and Opportunities. In M. D. C. VallsMartínez & J. Montero (Eds.), *Teaching Innovations in Economics.* Springer. https://doi.org/10.1007/978-3-031-72549-4_11

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS quarterly, 27*, 425–478. https://doi.org/10.2307/30036540

Wang, Y., Liu, C., & Tu, Y. F. (2021). Factors affecting the adoption of AI-based applications in higher education. *Educational Technology & Society, 24*(3), 116–129.

Wasserman, L. (2013). *All of statistics: A concise course in statistical inference*. Springer Science & Business Media.

Widyaningrum, R., Wulandari, F., Zainudin, M., Athiyallah, A., & Rizqa, M. (2024). Exploring the factors affecting ChatGPT acceptance among university students. *Multidisciplinary Science Journal, 6*(12), 2024273. https://doi.org/10.31893/multiscience.2024273

Widyanto, H. A., Kusumawardani, K. A., & Yohanes, H. (2022). Safety first: Extending UTAUT to better predict mobile payment adoption by incorporating perceived security, perceived risk, and trust. *Journal of Science and Technology Policy Management, 13*(4), 952–973. https://doi.org/10.1108/JSTPM-03-2020-0058

Wu, W., Zhang, B., Li, S., & Liu, H. (2022). Exploring factors of the willingness to accept AI-assisted learning environments: An empirical investigation based on the UTAUT model and perceived risk theory. *Frontiers in Psychology, 13*, 870777. https://doi.org/10.3389/fpsyg.2022.870777

Wu, J., Gan, W., Chen, Z., Wan, S., & Philip, S. Y. (2023). Multimodal large language models: A survey. *In 2023 IEEE International Conference on Big Data (BigData)*. IEEE, pp. 2247–2256. https://doi.org/10.1109/BigData59044.2023.10386743

Yakubu, M. N., & Dasuki, S. I. (2019). Factors affecting the adoption of e-learning technologies among higher education students in Nigeria: A structural equation modelling approach. *Information Development, 35*(3), 492–502. https://doi.org/10.1177/0266666918765907

Ylmaz, F. G., Yilmaz, R., & Ceylan, M. (2023). Generative artificial intelligence acceptance scale: A validity and reliability study. *International Journal of Human–Computer Interaction*, 1–13. https://doi.org/10.1080/10447318.2023.2288730

Zastudil, C., Rogalska, M., Kapp, C., Vaughn, J., & MacNeil, S. (2023). *Generative ai in computing education: Perspectives of students and instructors*. IEEE Frontiers in Education Conference (FIE), pp. 1–9.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Mohammed Nasiru Yakubu[1] · Nakama David[2] · Naima Hafiz Abubakar[3]

✉ Mohammed Nasiru Yakubu
nyakubu@arden.ac.uk

Nakama David
nakamadavid527@adsu.edu.ng

Naima Hafiz Abubakar
nhabubakar.it@buk.edu.ng

[1] Arden University, Coventry, UK

[2] Adamawa State University, Adamawa, Nigeria

[3] Bayero University, Kano, Nigeria